Time Series — Solutions

Practice Exam — Week 5: Trends, Seasonality & SARIMA

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Problem 1 (5 points)

Statement.

Consider the signal-plus-noise model $X_t = \mu_t + Y_t$, where μ_t is a deterministic trend and $\{Y_t\}$ is a zero-mean stationary process with autocovariance sequence γ_τ. Recall the difference operator $\nabla = I - B$ and the seasonal difference $\nabla_{(s)} = I - B^s$.

(a) *Take $\mu_t = a + bt$ (linear trend). Show that $\nabla X_t = b + \nabla Y_t$, and explain in one sentence why ∇Y_t is stationary while $\{X_t\}$ in general is not.* *(1.5 pts)*

(b) *Now let $\mu_t = a + bt + s_t$ with a* quarterly *season $s_{t+4} = s_t$. Compute $\nabla\nabla_{(4)} X_t$* explicitly *as a linear combination of the Y's, and verify that its mean is 0 and that it is weakly stationary.* *(2 pts)*

(c) (A classic trap.) *The operators $\nabla_{(s)} = I - B^s$ and $\nabla^s = (I - B)^s$ are* not *the same. For the deterministic sequence $\mu_t = t^2$, compute both $\nabla^2 \mu_t$ and $\nabla_{(2)} \mu_t$ and state which kind of structure each operator removes.* *(1.5 pts)*

Solution.

(a) With $\mu_t = a + bt$, linearity of $\nabla = I - B$ gives

$$\nabla X_t = X_t - X_{t-1} = \big(a + bt + Y_t\big) - \big(a + b(t-1) + Y_{t-1}\big) = b + \big(Y_t - Y_{t-1}\big) = b + \nabla Y_t.$$

So differencing collapses the linear trend $a + bt$ to the constant b. Now $\nabla Y_t = Y_t - Y_{t-1}$ is a *fixed finite linear combination* of a stationary process, hence itself stationary (its mean is 0 and its covariance depends on the lag only). The original $\{X_t\}$ is not stationary because $\mathbb{E}[X_t] = a + bt$ depends on t.

(b) With $\mu_t = a + bt + s_t$ and $s_{t+4} = s_t$, apply the seasonal difference first. Using $s_t = s_{t-4}$,

$$\nabla_{(4)} X_t = X_t - X_{t-4} = \big(a + bt + s_t + Y_t\big) - \big(a + b(t-4) + s_{t-4} + Y_{t-4}\big) = 4b + \big(Y_t - Y_{t-4}\big).$$

The period-4 season has been annihilated and the linear trend has collapsed to the constant $4b$. Now apply $\nabla = I - B$, which kills the constant:

$$\nabla\nabla_{(4)} X_t = \big(4b + Y_t - Y_{t-4}\big) - \big(4b + Y_{t-1} - Y_{t-5}\big) = \boxed{Y_t - Y_{t-1} - Y_{t-4} + Y_{t-5}}.$$

This is a fixed linear combination of $\{Y_t\}$, so its mean is $\mathbb{E}[Y_t - Y_{t-1} - Y_{t-4} + Y_{t-5}] = 0$, and by bilinearity its covariance,

$$\begin{aligned}\mathrm{Cov}\big(\nabla\nabla_{(4)} X_t,\ \nabla\nabla_{(4)} X_{t-\tau}\big) = {} & 4\gamma_\tau - 2\gamma_{\tau-1} - 2\gamma_{\tau+1} - 2\gamma_{\tau-4} - 2\gamma_{\tau+4} \\ & + \gamma_{\tau-3} + \gamma_{\tau+3} + \gamma_{\tau-5} + \gamma_{\tau+5} + \gamma_{\tau+1-4} + \gamma_{\tau-1+4},\end{aligned}$$

depends on the lag τ only (no t). Hence $\nabla\nabla_{(4)} X_t$ is weakly stationary; all t-dependence has been removed. (One need not simplify the covariance fully; the key point is that it is a function of τ alone.)

(c) Treat $\mu_t = t^2$ as a deterministic sequence.

Ordinary second difference $\nabla^2 = (I - B)^2$:

$$\nabla^2 t^2 = t^2 - 2(t-1)^2 + (t-2)^2 = t^2 - 2\big(t^2 - 2t + 1\big) + \big(t^2 - 4t + 4\big) = 2,$$

a constant. (And $\nabla^3 t^2 = \nabla(2) = 0$.) So ∇^q *annihilates a polynomial trend of degree* $q-1$: here ∇^2 reduces the degree-2 polynomial t^2 to a degree-0 constant, and one further ∇ kills it.

Seasonal difference of period 2, $\nabla_{(2)} = I - B^2$:

$$\nabla_{(2)} t^2 = t^2 - (t-2)^2 = t^2 - \big(t^2 - 4t + 4\big) = 4t - 4,$$

which is still linear (not annihilated). The seasonal operator $\nabla_{(s)}$ removes a deterministic *period-s season* ($\nabla_{(s)} s_t = s_t - s_{t-s} = 0$), *not* a polynomial trend. The two operators serve different purposes, and $\nabla_{(2)} \neq \nabla^2$.

Problem 2 (5 points)

Statement.

Let $\{X_t\}$ *be the multiplicative seasonal process* $\mathrm{SARMA}(0,1) \times (1,0)_4$ *(quarterly data), defined by*

$$\big(1 - \Phi\, B^4\big) X_t = \big(1 - \theta B\big)\varepsilon_t, \qquad |\Phi| < 1,$$

a seasonal autoregression at lag 4 *driven by a non-seasonal* MA(1) *input.*

(a) *Write the explicit one-step recursion for* X_t*, and by inverting the seasonal AR factor obtain the* $\mathrm{MA}(\infty)$ *weights* ψ_j *in* $X_t = \sum_{j\geq 0} \psi_j \varepsilon_{t-j}$ *(give a closed form for* ψ_{4l} *and* ψ_{4l+1}*, and state that all other* $\psi_j = 0$*). (2 pts)*

(b) *Using the white-noise filter rule* $\gamma_\tau = \sigma^2 \sum_{j\geq 0} \psi_j \psi_{j+\tau}$*, compute* $\gamma_0, \gamma_1, \gamma_3, \gamma_4, \gamma_5$ *in closed form, and explain why the autocovariance lives in clusters around the seasonal lags* $0, 4, 8, \ldots$ *(2 pts)*

(c) *Evaluate the autocorrelations* ρ_1, ρ_3, ρ_4 *numerically for* $\Phi = \frac{1}{2}$, $\theta = \frac{2}{5}$, $\sigma^2 = 1$. *(1 pt)*

Solution.

(a) Expand the seasonal AR factor $(1 - \Phi B^4) X_t = (1 - \theta B)\varepsilon_t$ and solve for X_t:

$$\boxed{X_t = \Phi\, X_{t-4} + \varepsilon_t - \theta \varepsilon_{t-1}}$$

— an AR(1) acting only on the seasonal lag 4, driven by an MA(1) input. Invert the seasonal factor as a geometric series in B^4 (valid since $|\Phi| < 1$),

$$\frac{1}{1 - \Phi B^4} = \sum_{l \geq 0} \Phi^l B^{4l},$$

so that

$$X_t = \sum_{l\geq 0} \Phi^l B^{4l} \big(\varepsilon_t - \theta \varepsilon_{t-1}\big) = \sum_{l \geq 0} \Phi^l \big(\varepsilon_{t-4l} - \theta \varepsilon_{t-4l-1}\big).$$

Reading off the coefficient of ε_{t-j}:

$$\boxed{\psi_{4l} = \Phi^l, \qquad \psi_{4l+1} = -\theta\, \Phi^l \quad (l \geq 0), \qquad \psi_j = 0 \text{ otherwise.}}$$

(Check: $\psi_0 = 1,\ \psi_1 = -\theta,\ \psi_4 = \Phi,\ \psi_5 = -\theta\Phi, \ldots$, and indeed $\psi_j - \Phi \psi_{j-4}$ reproduces the MA(1) numerator $1 - \theta B$.)

(b) Apply $\gamma_\tau = \sigma^2 \sum_{j \geq 0} \psi_j \psi_{j+\tau}$. The only nonzero taps sit at $j \equiv 0$ and $j \equiv 1 \pmod 4$, so a product $\psi_j \psi_{j+\tau}$ survives only when both indices are of this type; this happens at $\tau \in \{0, \pm 1, \pm 3, \pm 4, \pm 5, \ldots\}$, i.e. in clusters $\{4k-1,\ 4k,\ 4k+1\}$ around the seasonal multiples $4k$.

Lag 0*:* $\sum_l \psi_{4l}^2 + \sum_l \psi_{4l+1}^2 = \sum_l \Phi^{2l} + \theta^2 \sum_l \Phi^{2l} = (1+\theta^2)/(1-\Phi^2)$, so

$$\gamma_0 = \frac{(1+\theta^2)\,\sigma^2}{1 - \Phi^2}.$$

Lag 1*:* the surviving pairs are $\psi_{4l}\psi_{4l+1} = \Phi^l(-\theta\Phi^l) = -\theta\Phi^{2l}$, so

$$\gamma_1 = \sigma^2 \sum_{l\geq 0} \bigl(-\theta\Phi^{2l}\bigr) = \frac{-\theta\,\sigma^2}{1-\Phi^2}.$$

Lag 3*:* pairs $\psi_{4l+1}\psi_{4l+4} = (-\theta\Phi^l)(\Phi^{l+1}) = -\theta\Phi^{2l+1}$, so

$$\gamma_3 = \sigma^2 \sum_{l\geq 0} \bigl(-\theta\Phi^{2l+1}\bigr) = \frac{-\theta\,\Phi\,\sigma^2}{1-\Phi^2}.$$

Lag 4*:* pairs $\psi_{4l}\psi_{4l+4} = \Phi^{2l+1}$ and $\psi_{4l+1}\psi_{4l+5} = (-\theta\Phi^l)(-\theta\Phi^{l+1}) = \theta^2\Phi^{2l+1}$, so

$$\gamma_4 = \sigma^2 \sum_{l\geq 0} (1+\theta^2)\Phi^{2l+1} = \frac{(1+\theta^2)\Phi\,\sigma^2}{1-\Phi^2} = \Phi\,\gamma_0.$$

Lag 5*:* pairs $\psi_{4l}\psi_{4l+5} = \Phi^l(-\theta\Phi^{l+1}) = -\theta\Phi^{2l+1}$, so

$$\gamma_5 = \frac{-\theta\,\Phi\,\sigma^2}{1-\Phi^2} = \gamma_3.$$

Collecting,

$$\boxed{\gamma_0 = \frac{(1+\theta^2)\sigma^2}{1-\Phi^2}, \quad \gamma_1 = \frac{-\theta\,\sigma^2}{1-\Phi^2}, \quad \gamma_3 = \gamma_5 = \frac{-\theta\Phi\,\sigma^2}{1-\Phi^2}, \quad \gamma_4 = \Phi\,\gamma_0.}$$

So the seasonal AR(1) produces a spike at every seasonal lag $4k$ of size $\Phi^k\gamma_0$, each flanked by MA(1) side-lobes at $4k \pm 1$; the whole pattern decays geometrically in Φ.

(c) With $\Phi = \frac{1}{2}$, $\theta = \frac{2}{5}$, $\sigma^2 = 1$: $1 - \Phi^2 = \frac{3}{4}$ and $1 + \theta^2 = 1 + \frac{4}{25} = \frac{29}{25}$. Hence

$$\gamma_0 = \frac{29/25}{3/4} = \frac{116}{75} \approx 1.5467.$$

Then

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{-\theta}{1+\theta^2} = \frac{-2/5}{29/25} = \frac{-2/5 \cdot 25}{29} = -\frac{10}{29} \approx -0.3448,$$

$$\rho_4 = \frac{\gamma_4}{\gamma_0} = \Phi = \tfrac{1}{2} = 0.5, \qquad \rho_3 = \frac{\gamma_3}{\gamma_0} = \Phi\,\rho_1 = \tfrac{1}{2} \cdot \left(-\tfrac{10}{29}\right) = -\frac{5}{29} \approx -0.1724.$$

(So the correlogram shows a dominant seasonal spike $\rho_4 = 0.5$ with small negative side-lobes $\rho_3 = \rho_5 \approx -0.172$ at 4 ± 1, and $\rho_1 \approx -0.345$ near the origin.)

Problem 3 (5 points)

Statement.

This problem studies non-stationary integrated models.

(a) (Random walk.) *Let $X_t = \sum_{j=1}^{t} \varepsilon_j$ with $X_0 = 0$ (an* ARIMA(0,1,0)*). Compute* $\mathbb{E}[X_t]$, $\mathrm{Var}(X_t)$ *and* $\mathrm{Cov}(X_t, X_{t+\tau})$ *for $\tau \geq 0$, deduce that $\{X_t\}$ is* not *stationary, and show ∇X_t is white noise. Show also that* $\mathrm{Corr}(X_t, X_{t+\tau}) \to 1$ *as $t \to \infty$ with τ fixed.* *(2 pts)*

(b) (IMA(1,1).) *Let $Y_t = Y_{t-1} + \varepsilon_t - \theta\varepsilon_{t-1}$ with $Y_0 = 0$. By unrolling the recursion, show that for $t \geq 1$*

$$Y_t = \varepsilon_t + (1-\theta)\sum_{j=1}^{t-1} \varepsilon_j - \theta\varepsilon_0,$$

and hence compute $\mathrm{Var}(Y_t)$*. For $\theta \neq 1$, what is the order of growth of* $\mathrm{Var}(Y_t)$ *in t?* *(2 pts)*

(c) (Over-differencing.) *What happens to* Y_t *and to* $\mathrm{Var}(Y_t)$ *at the boundary* $\theta = 1$*? Show that* ∇Y_t *is then a* non-invertible MA(1), *and state the value of its lag-*1 *autocorrelation* ρ_1. *Why is this the warning signature of over-differencing?* *(1 pt)*

Solution.

(a) Since $X_t = \sum_{j=1}^{t} \varepsilon_j$ is a sum of mean-zero noise, $\mathbb{E}[X_t] = 0$. As the ε_j are uncorrelated with variance σ^2,

$$\mathrm{Var}(X_t) = \sum_{j=1}^{t} \mathrm{Var}(\varepsilon_j) = t\,\sigma^2.$$

For $\tau \geq 0$, $X_{t+\tau} = X_t + \sum_{j=t+1}^{t+\tau} \varepsilon_j$ adds only *new* (uncorrelated) noise, so the overlap is the first t shared terms:

$$\mathrm{Cov}(X_t, X_{t+\tau}) = \mathrm{Cov}\Big(\sum_{j=1}^{t} \varepsilon_j,\ \sum_{k=1}^{t+\tau} \varepsilon_k\Big) = \sum_{j=1}^{t} \sigma^2 = t\,\sigma^2.$$

Both $\mathrm{Var}(X_t)$ and $\mathrm{Cov}(X_t, X_{t+\tau})$ depend on t, so $\{X_t\}$ is **not** (weakly) stationary. However

$$\nabla X_t = X_t - X_{t-1} = \varepsilon_t,$$

which is white noise; hence $\{X_t\}$ is an $\mathrm{ARIMA}(0, 1, 0)$. Finally

$$\mathrm{Corr}(X_t, X_{t+\tau}) = \frac{t\sigma^2}{\sqrt{t\sigma^2}\sqrt{(t+\tau)\sigma^2}} = \sqrt{\frac{t}{t+\tau}} \xrightarrow[t\to\infty]{} 1 \quad (\tau \text{ fixed}),$$

the near-unit correlation of nearby values characteristic of an integrated series.

(b) The recursion $Y_t = Y_{t-1} + (\varepsilon_t - \theta\varepsilon_{t-1})$ with $Y_0 = 0$ telescopes:

$$Y_t = \sum_{k=1}^{t} (\varepsilon_k - \theta\varepsilon_{k-1}) = \sum_{k=1}^{t} \varepsilon_k - \theta \sum_{k=1}^{t} \varepsilon_{k-1} = \sum_{k=1}^{t} \varepsilon_k - \theta \sum_{j=0}^{t-1} \varepsilon_j.$$

Separating the unmatched endpoints ε_t (only in the first sum) and ε_0 (only in the second),

$$Y_t = \varepsilon_t + \sum_{k=1}^{t-1} \varepsilon_k - \theta\varepsilon_0 - \theta \sum_{j=1}^{t-1} \varepsilon_j = \varepsilon_t + (1-\theta) \sum_{j=1}^{t-1} \varepsilon_j - \theta\varepsilon_0,$$

as claimed. The coefficients of the $t+1$ independent innovations $\varepsilon_0, \varepsilon_1, \ldots, \varepsilon_t$ are: $-\theta$ on ε_0, $(1-\theta)$ on each of $\varepsilon_1, \ldots, \varepsilon_{t-1}$ (that is $t-1$ terms), and 1 on ε_t. By independence,

$$\boxed{\mathrm{Var}(Y_t) = \sigma^2 \Big[1 + (1-\theta)^2 (t-1) + \theta^2 \Big].}$$

For $\theta \neq 1$ this is *linear in* t with positive slope $(1-\theta)^2\sigma^2 > 0$: $\mathrm{Var}(Y_t) = O(t)$, exactly the random-walk-like variance growth that flags an integrated (needs-differencing) series.

(c) At the boundary $\theta = 1$ the slope $(1-\theta)^2$ vanishes, so

$$\mathrm{Var}(Y_t) = \sigma^2 \big[1 + 0 + 1\big] = 2\sigma^2,$$

which is *bounded* (no growth). Indeed the recursion telescopes completely:

$$Y_t = \sum_{k=1}^{t} (\varepsilon_k - \varepsilon_{k-1}) = \varepsilon_t - \varepsilon_0,$$

so the level is itself stationary, and

$$\nabla Y_t = Y_t - Y_{t-1} = \varepsilon_t - \varepsilon_{t-1},$$

an MA(1) with unit coefficient $\theta = 1$. Its MA polynomial $1 - z$ has root $z = 1$ *on* the unit circle, so it is **non-invertible**, and its lag-1 autocorrelation is

$$\rho_1 = \frac{-\theta}{1+\theta^2} = \frac{-1}{1+1} = -\tfrac{1}{2}.$$

This is the warning signature of **over-differencing**: differencing a series that was already stationary injects a $(1 - B)$ factor into the MA part, i.e. a non-invertible MA(1) with $\rho_1 = -\frac{1}{2}$. Seeing a sample $\rho_1 \approx -\frac{1}{2}$ after a difference is the cue to difference one time fewer.

Problem 4 (5 points)

Statement.

(Revision of Week 4: Yule–Walker estimation.) *Let $\{X_t\}$ be a mean-zero causal* AR(2) *process $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ observed at $X_1, \ldots, X_n$, with sample autocovariances*

$$\hat{\gamma}_0 = 10, \qquad \hat{\gamma}_1 = 5, \qquad \hat{\gamma}_2 = 1.$$

(a) *Derive the Yule–Walker equations $\gamma_k = \phi_1\gamma_{k-1} + \phi_2\gamma_{k-2}$ $(k \geq 1)$ and the variance identity $\gamma_0 = \phi_1\gamma_1 + \phi_2\gamma_2 + \sigma^2$, making clear where* causality *is used.* *(2 pts)*

(b) *Set up the 2×2 Yule–Walker system, invert it* by hand*, and report the estimates $\hat{\phi}_1, \hat{\phi}_2$ and $\hat{\sigma}^2$.* *(2 pts)*

(c) *Verify that the fitted model is causal (stationary), e.g. by locating the roots of $\hat{\Phi}(z)$ or via the* AR(2) *stationarity triangle.* *(1 pt)*

Solution.

(a) Multiply the model $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ by X_{t-k} and take expectations. The process is causal, so X_{t-k} is a one-sided combination of $\varepsilon_{t-k}, \varepsilon_{t-k-1}, \ldots$; hence for $k \geq 1$ the noise term is uncorrelated with the past, $\mathbb{E}[\varepsilon_t X_{t-k}] = 0$. With mean zero, $\mathbb{E}[X_{t-j}X_{t-k}] = \gamma_{k-j}$, giving for $k \geq 1$

$$\gamma_k = \phi_1\gamma_{k-1} + \phi_2\gamma_{k-2} \qquad (\text{using } \gamma_{-1} = \gamma_1).$$

Multiplying instead by X_t keeps the noise term: by causality and $\psi_0 = 1$,

$$\mathbb{E}[\varepsilon_t X_t] = \mathbb{E}\big[\varepsilon_t(\phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t)\big] = \sigma^2,$$

so the variance identity is

$$\gamma_0 = \phi_1\gamma_1 + \phi_2\gamma_2 + \sigma^2.$$

The single fact "causality $\Rightarrow \mathbb{E}[\varepsilon_t X_{t-k}] = 0$ for $k \geq 1$" is what closes the system. Replacing the population γ by the sample $\hat{\gamma}$ gives the Yule–Walker *estimators*.

(b) Taking $k = 1, 2$ gives the 2×2 Yule–Walker system $\hat{\Gamma}_2\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\gamma}}_2$:

$$\begin{pmatrix} \hat{\gamma}_0 & \hat{\gamma}_1 \\ \hat{\gamma}_1 & \hat{\gamma}_0 \end{pmatrix} \begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} = \begin{pmatrix} \hat{\gamma}_1 \\ \hat{\gamma}_2 \end{pmatrix} \quad \Longrightarrow \quad \begin{pmatrix} 10 & 5 \\ 5 & 10 \end{pmatrix} \begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

The inverse of $\left(\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right)$ is $\frac{1}{a^2-b^2}\left(\begin{smallmatrix} a & -b \\ -b & a \end{smallmatrix}\right)$; here $a = 10,\ b = 5$ and $a^2 - b^2 = 100 - 25 = 75$. Hence

$$\begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} = \frac{1}{75}\begin{pmatrix} 10 & -5 \\ -5 & 10 \end{pmatrix}\begin{pmatrix} 5 \\ 1 \end{pmatrix} = \frac{1}{75}\begin{pmatrix} 10 \cdot 5 - 5 \cdot 1 \\ -5 \cdot 5 + 10 \cdot 1 \end{pmatrix} = \frac{1}{75}\begin{pmatrix} 45 \\ -15 \end{pmatrix} = \begin{pmatrix} 0.6 \\ -0.2 \end{pmatrix}.$$

The innovation variance is

$$\hat{\sigma}^2 = \hat{\gamma}_0 - \hat{\phi}_1\hat{\gamma}_1 - \hat{\phi}_2\hat{\gamma}_2 = 10 - 0.6 \cdot 5 - (-0.2) \cdot 1 = 10 - 3 + 0.2 = 7.2.$$

So $\boxed{\hat{\phi}_1 = 0.6,\ \hat{\phi}_2 = -0.2,\ \hat{\sigma}^2 = 7.2.}$ (Consistency check via $\gamma_0 = \sigma^2/(1 - \phi_1\rho_1 - \phi_2\rho_2)$ with $\rho_1 = \frac{1}{2}, \rho_2 = \frac{1}{10}$: $1 - 0.6 \cdot \frac{1}{2} - (-0.2) \cdot \frac{1}{10} = 1 - 0.3 + 0.02 = 0.72$, and $7.2/0.72 = 10 = \hat{\gamma}_0$. ✓)

(c) The fitted AR polynomial is $\hat{\Phi}(z) = 1 - 0.6z + 0.2z^2$. Its roots solve $0.2z^2 - 0.6z + 1 = 0$, i.e. $z^2 - 3z + 5 = 0$:

$$z = \frac{3 \pm \sqrt{9 - 20}}{2} = \frac{3 \pm \sqrt{-11}}{2} = \tfrac{3}{2} \pm \tfrac{\sqrt{11}}{2}\, i, \qquad |z| = \sqrt{\left(\tfrac{3}{2}\right)^2 + \tfrac{11}{4}} = \sqrt{\tfrac{9}{4} + \tfrac{11}{4}} = \sqrt{5} \approx 2.236.$$

Both roots have modulus $\sqrt{5} > 1$ (outside the unit circle), so the fit is **causal/stationary**. Equivalently, the AR(2) stationarity-triangle conditions hold: $\hat{\phi}_1 + \hat{\phi}_2 = 0.4 < 1$, $\hat{\phi}_2 - \hat{\phi}_1 = -0.8 < 1$, and $\left|\hat{\phi}_2\right| = 0.2 < 1$. (Unlike least squares, the Yule–Walker estimator always returns a stationary fit.)